

The proper detail if we go with proper industrial Device

(Actual Device)

Oil Device Gas Data Extraction:

Our Dissolved Gas Analysis (DGA) device is a smart, affordable tool to detect transformer faults early by extracting oil, releasing hidden gases, and measuring them. It works on live (hot) or idle (cool) transformers no shutdown needed. We pull a small oil sample, filter it, pump it through a heated degasser to "pop out" gases (using Henry's Law: warmer oil releases gases better), sense the gases, analyze for faults (like sparks or overheating), and return the oil safely.

Proper Flow (Step-by-Step):

1. **Connect & Extract:** Attach to transformer valve → Open valve → Pump pulls oil (100–200 mL).
2. **Clean & Heat:** Filter dirt → Heat to 70°C if cool (for best gas release).
3. **Release Gases:** In degasser chamber shake/vacuum oil → Gases bubble out.
4. **Measure:** Gases go to sensor chamber → Sensors read values.
5. **Finish Up:** Vent extra gases → Return oil → Controller runs algo → Show results/alerts.

Controller:

- Controller: ESP32 or Raspberry Pi Zero
- Communications/UI: WiFi/Bluetooth module + 2.8" TFT LCD screen

Equipment Required:

Equipment	Recommended Type/Model	Why We Need It
Oil Tap Connector	1/4" NPT brass adapter + quick-connect fitting	Safely hooks to transformer's sampling valve (no leaks on hot oil).
Sample Valve	12V solenoid valve (stainless steel, 100°C rated)	Auto-opens/closes oil flow for safe extraction.

Particulate Filter	Inline 10-micron mesh filter (stainless)	Removes dirt/sludge to protect pump and chambers.
Flow Pump	12V peristaltic pump (0.2–1 L/min, e.g., Kamoer or generic)	Gently pulls and circulates oil without contamination. Handles 100°C oil.
Degasser Chamber	200 mL glass vacuum flask + shaker motor (or membrane like PDMS sheet)	Shakes/vacuums oil to release gases into headspace.
Heater	50W PID heating coil + controller (e.g., Inkbird ITC-100)	Warms oil to 70°C if cool or boosts gas release (Henry's Law).
Gas Sensor Chamber	Small sealed acrylic box (100 mL, with inlet/outlet ports)	Holds sensors; collects pure gases for accurate reading.
Gas Vent/Scrubber	Exhaust valve + activated carbon filter	Safely releases extra gases (no buildup or smell).
Oil Return Tubing	1/4" silicone hose (5m, 150°C rated) + fittings	Sends clean oil back to transformer.
Enclosure Case	IP65 waterproof plastic box (30x20x10 cm)	Houses protect everything from dust/weather.
Power Supply	12V/24V DC adapter + rechargeable battery (e.g., 5000mAh)	Powers pump, heaters, sensors portable for field use.

Sensor required for detect values:

Sensor	Recommended Type/Model	What It Detects
Main Gas Sensors (Array)	- H ₂ : Alphasense H2-A1 or MQ-8 (0–2000 ppm) - CO/CH ₄ : SPEC Sensors IR or Figaro TGS - Multi-gas: Winsen ZE08 (for C ₂ H ₄ /C ₂ H ₂)	Key fault gases: H ₂ (sparks), CH ₄ /CO (heat), etc.
Temperature Sensor	DS18B20 waterproof probe (-50 to 150°C)	Oil temp in flow.
Pressure Sensor	MPX5700 or BMP280 (0–10 bar)	System pressure.
Moisture Sensor	Capacitive oil humidity probe (e.g., HL-1)	Water in oil.